STATISTICS · STATISTICAL INFERENCE

Confidence Interval Lab

Explore & Reflect **TEACHER KEY**

Name _____ Date _____ Period _____

Open **Confidence Interval Lab** in your unit's Interactive Math section on Mr. K's Classroom (or use your teacher's link). Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3. Do Parts 4 and 5 on your own.

1 Predict before you touch the tool

1. If you make 100 different 95% confidence intervals from 100 random samples, how many do you expect to capture the true value?
   About 95, but the count varies from run to run.
2. Is a 99% interval from the same sample wider or narrower than a 95% interval? Why?
   Wider: more confidence requires a larger critical value (2.576 instead of 1.960).

2 Explore use the tool

1. Proportion, p = 0.4, n = 40, 95%: make 100 or more intervals. Record how many you made and how many captured p.
   Answers vary; usually about 92%–97% capture.
2. Same n: record the average margin of error at 90% and at 99%.
   About 0.127 at 90% and 0.200 at 99% (theory with $\hat{p} = 0.4$).
3. At 95%, what n cuts the average margin of error from n = 40 roughly in half?
   n = 160 (4 × 40): about 0.152 → 0.076.
4. Mean, n = 5: make 200+ intervals with z* and s, then 200+ with t*. Record both capture rates and explain the difference.
   About 88% with z* and about 95% with t*. t* = 2.776 (df = 4) is larger, which accounts for the extra uncertainty from estimating σ with s.

3 Record use the tool and fill in the table

Margins of error

Use the formulas shown in the tool's readout.

Setting	Critical value	Standard error	Margin of error
$\hat{p}$ = 0.4, n = 40, 95%	z* = 1.960	√(0.4·0.6/40) ≈ 0.0775	≈ 0.152
$\hat{p}$ = 0.4, n = 160, 95%	z* = 1.960	≈ 0.0387	≈ 0.076
$\hat{p}$ = 0.4, n = 40, 99%	z* = 2.576	≈ 0.0775	≈ 0.200
$\bar{x}$, s = 10, n = 5, 95%	t* = 2.776 (df 4)	10/√5 ≈ 4.472	≈ 12.42
$\bar{x}$, s = 10, n = 30, 95%	t* = 2.045 (df 29)	10/√30 ≈ 1.826	≈ 3.734

4 Explain what did you discover?

1. "95% confident" means that about 95% of intervals from repeated random samples (the method) capture the parameter.
2. Margin of error = critical value × standard error .

3. To halve the margin of error, use about 4 times the sample size.

4. When σ is unknown, use t* with df = n – 1.

5. From one sample, Ana computes the 95% interval (0.31, 0.47) and says, "There is a 95% probability that p is between 0.31 and 0.47." What is wrong? Write a correct interpretation.

   p is fixed, so this interval either captures it or not. Correct: "We are 95% confident that the interval from 0.31 to 0.47 captures the true proportion p," where 95% is the long-run capture rate of the method.

5 Apply a new situation, no tool

1. In an SRS of 500 voters, 280 support a measure. Construct a 95% confidence interval for p. (Conditions: 280 and 220 are both ≥ 10.)

   $\hat{p} = 0.56$, SE ≈ 0.0222, ME ≈ 0.0435: (0.5165, 0.6035).

2. How large a sample gives a margin of error of at most 0.03 at 95% confidence with no prior estimate of p?

   $n \geq (1.96/0.03)^2 \times 0.25 \approx 1067.1$, so n = 1068.

3. An SRS of 12 bottles has $\bar{x}$ = 501.2 mL and s = 2.4 mL. Construct a 95% confidence interval for μ.

   t* = 2.201 (df 11), SE ≈ 0.6928, ME ≈ 1.525: (499.68, 502.72) mL.

Teacher notes

Intended discovery: A confidence level describes the long-run capture rate of the method; higher confidence widens intervals, quadrupling n halves the margin of error, and capture rates fall short when conditions fail or z* is used with s for small n.

Likely misconceptions:

- "95% probability that p is in this interval" (Explain).
- Thinking a larger sample raises the confidence level (it narrows the interval).
- Using z* with s for small samples.

Answers that can vary: All simulated capture counts and averages (Explore 1–4).